From Novel View Synthesis to Generative World Modeling: A Comprehensive Survey

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Abstract

Novel View Synthesis (NVS)—the task of rendering a scene from arbitrary unseen viewpoints given a set of reference observations—has undergone a paradigm shift from classical geometry-based reconstruction toward learning-driven generative approaches. This survey examines NVS not merely as a rendering technique, but as the **foundational spatial capability** of Generative World Models: systems that predict visual scene states under arbitrary conditions, including viewpoint changes, temporal dynamics, and agent interactions. We present a unified conceptual framework that organises existing methods along a *continuous spectrum*—from static multi-view reconstruction to dynamic, interactive world generation—across three axes: (i) scene representation (implicit vs. explicit vs. hybrid), (ii) generative paradigm (per-scene optimisation, feed-forward inference, and diffusion-based synthesis), and (iii) world-modelling capability (geometry, dynamics, and controllability). We survey more than 150 representative methods, identify the key open challenges, and lay out a roadmap toward physically grounded, generalisable world models.

1 Introduction

1.1 The Fundamental Challenge

Intelligence requires reasoning about the world beyond what is immediately visible. A self-driving vehicle must anticipate the scene around a blind corner; a robot must predict how a grasped object will appear from the wrist camera; a surgeon must mentally rotate an anatomy from a 2-D scan. In each case, the core operation is identical: **synthesising a plausible visual state at an unobserved vantage point**.

This is the problem of **Novel View Synthesis (NVS)**. Given reference images $\{\mathbf{I}_i, \boldsymbol{\pi}_i\}_{i=1}^n$ at known camera poses, the goal is to render a photo-realistic image $\hat{\mathbf{I}}$ at an arbitrary target pose $\boldsymbol{\pi}^*$, faithfully preserving geometry, appearance, and lighting.

While the problem was studied in the image-based rendering era [1, 2], the past five years have witnessed an extraordinary leap: NeRF [12], 3DGS [27], and diffusion-model-based methods [41, 44] have progressively pushed the frontier from slow per-scene reconstruction toward fast, generalisable, and generative synthesis.

1.2 From NVS to World Modeling

We argue that this progression is not merely a technical improvement—it reflects a deeper conceptual shift: **NVS is becoming the spatial inference engine of Generative World Modeling**.

Preprint.

Table 1: Comparison with related surveys. + covered; ∼ partially covered; – not covered.

Survey	Year	NeRF	3DGS	Generative	Dynamic	Generalise	4D/WM	Unified
Tewari *et al.* [5]	2022	+	–	∼	∼	∼	–	–
Gao *et al.* [6]	2022	+	–	–	∼	–	–	–
Zhu *et al.* [7]	2023	∼	–	–	–	–	–	–
Wu *et al.* [8]	2024	–	+	–	∼	–	–	–
Zhang *et al.* [10]	2025	∼	∼	–	–	+	–	–
He *et al.* [11]	2025	–	+	∼	∼	–	–	–
Ours	2026	+	+	+	+	+	+	+

Working Definition (World Model). A *Generative World Model* is a learned system that can predict the visual (and optionally physical) state of a scene under arbitrary conditions—viewpoint changes, temporal evolution, and agent interactions—in a manner that is *geometrically consistent*, *physically plausible*, and *generalisable* beyond the training distribution.

Under this definition, NVS is the special case where the only varying condition is camera pose. A world model must additionally handle temporal dynamics and action conditioning. Crucially, *spatial consistency enforced by NVS is a non-negotiable prerequisite*: a world model without geometric grounding is merely a video generator—plausible in appearance but unreliable for downstream spatial reasoning.

1.3 Relationship to Prior Surveys

Table 1 compares this survey with the most closely related prior works. Prior surveys on NeRF [5, 6] and 3DGS [8, 9] focus on *single representation paradigms* and do not address the connection to world modelling. The survey by Zhu *et al.* [7] covers image-based rendering broadly but predates the diffusion and 4-D modelling era. Zhang *et al.* [10] survey feed-forward reconstruction exclusively, omitting generative and world-modelling perspectives. The survey by He *et al.* [11] covers 3DGS applications but is not focused on NVS as a unified problem. To our knowledge, **no existing survey jointly covers all three paradigms (NeRF, 3DGS, and generative methods) through the unifying lens of Generative World Modeling**, nor proposes a continuous-spectrum taxonomy from static reconstruction to 4-D interactive generation.

1.4 The Unified Spectrum

We organise all methods along a *continuous spectrum* defined by three axes (Figure 1):

Axis 1. Input regime: dense multi-view → sparse-view → single image → text/unconditional.

Axis 2. Temporal scope: static scene → deformable/dynamic → 4-D world generation.

Axis 3. Controllability: viewpoint only → viewpoint + time → viewpoint + time + action.

Methods in the upper-left of this spectrum (dense input, static, viewpoint only) are classical reconstruction methods; those in the lower-right (minimal input, 4-D, action-conditioned) are Generative World Models. The trajectory from one end to the other is precisely the history of NVS.

1.5 Contributions

(1) **Unified Framework.** We introduce a continuous-spectrum taxonomy with three axes (input, temporal, controllability), providing the first unified organisation of NVS methods from static reconstruction to 4-D world generation.

(2) **Cross-Paradigm Coverage.** We survey >150 methods across NeRF, 3DGS, and generative paradigms, with explicit world-modelling implications for each family.

(3) **Related Survey Gap Analysis.** Table 1 demonstrates a clear gap: no existing survey covers all paradigms with a world-modelling lens.

(4) **Open Problems and Roadmap.** We identify the geometry–generation trade-off, long-horizon coherence, physical grounding, and evaluation as the four critical open challenges.

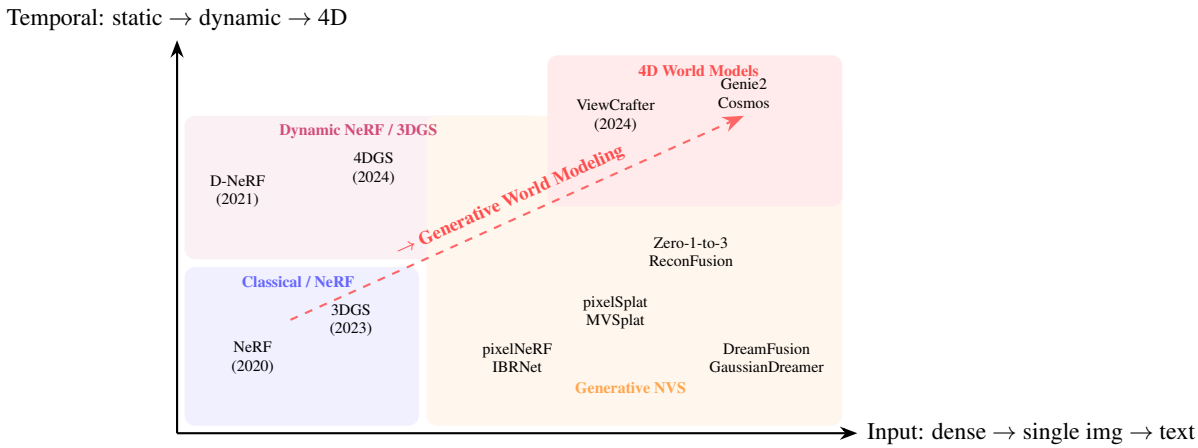

Figure 1: The unified NVS spectrum. Methods are positioned by input regime (x-axis) and temporal scope (y-axis). The diagonal arrow marks the evolutionary trajectory from per-scene static reconstruction toward 4-D Generative World Modeling.

1.6 Paper Structure

§2 Background. §3 NeRF. §**??** 3DGS. §4 Generative NVS. §5 Towards World Modeling. §6 Datasets. §7 Evaluation & Limitations. §8 Open Problems. §9 Conclusion.

2 Background and Unified Formulation

2.1 Camera Model

We adopt the standard pinhole camera model. A 3-D point $\mathbf{X} \in \mathbb{R}^3$ projects to pixel $\mathbf{p} \in \mathbb{R}^2$ via $\mathbf{p} = \mathbf{K}[\mathbf{R}|\mathbf{t}]\,\tilde{\mathbf{X}}$, where $\mathbf{K}$ is the intrinsic matrix and $(\mathbf{R}, \mathbf{t})$ the extrinsic pose. A camera ray emitted from centre $\mathbf{o}$ along direction $\mathbf{d}$ is $\mathbf{r}(s) = \mathbf{o} + s\mathbf{d}$, $s \in [s_n, s_f]$.

2.2 Volume Rendering

NeRF [12] represents a scene as $F_\theta : (\mathbf{x}, \mathbf{d}) \mapsto (\mathbf{c}, \sigma)$. The expected colour of a ray is

$$\hat{C}(\mathbf{r}) = \int_{s_n}^{s_f} T(s)\,\sigma(\mathbf{r}(s))\,\mathbf{c}(\mathbf{r}(s), \mathbf{d})\,\mathrm{d}s, \quad T(s) = \exp\left(-\int_{s_n}^{s} \sigma(\mathbf{r}(u))\,\mathrm{d}u\right). \tag{1}$$

2.3 Differentiable Rasterisation

3DGS [27] represents a scene as N anisotropic Gaussians $\{\mu_i, \Sigma_i, \alpha_i, \mathbf{c}_i\}$. Pixel colour is computed via front-to-back alpha compositing:

$$\hat{C} = \sum_{i \in \mathcal{N}} \mathbf{c}_i \alpha_i \tilde{G}_i \prod_{j<i} (1 - \alpha_j \tilde{G}_j), \tag{2}$$

where $\tilde{G}_i$ is the 2-D projection of Gaussian i.

2.4 Unified NVS Formulation

We formulate NVS as:

$$\mathcal{M}_\theta : \left(\{\mathbf{I}_i, \boldsymbol{\pi}_i\}_{i=1}^{n},\ \boldsymbol{\pi}^*\right) \longrightarrow \hat{\mathbf{I}}^*, \tag{3}$$

where n ranges from dense ($n \gg 1$) to single-image ($n = 1$). The world-modelling extension adds time t and action $\mathbf{a}$:

$$\mathcal{M}_\theta : \left(\{\mathbf{I}_i, \boldsymbol{\pi}_i\}_{i=1}^{n},\ \mathbf{a}_{0:t},\ t,\ \boldsymbol{\pi}^*\right) \longrightarrow \hat{\mathbf{I}}_t^*. \tag{4}$$

Eqs. 3–4 define five desiderata: **D1** geometric consistency; **D2** sparse/single-image generalisation; **D3** dynamic scene modelling; **D4** action conditioning; **D5** physical plausibility.

Table 2: Representative NeRF variants and their properties.

Method	Venue	Train Speed	Render Speed	Generalise	Dynamic	Key Idea
NeRF [12]	ECCV'20	Slow (hrs)	Slow	–	–	MLP + volume rendering
Instant-NGP [15]	TOG'22	Fast (sec)	Fast	–	–	Multi-res hash grid
TensoRF [16]	ECCV'22	Med (min)	Fast	–	–	Tensor decomposition
Plenoxels [17]	CVPR'22	Fast (min)	Fast	–	–	Voxel grid
Mip-NeRF [18]	ICCV'21	Slow	Slow	–	–	Anti-aliasing
Mip-NeRF 360 [19]	CVPR'22	Slow	Slow	–	–	Unbounded scenes
pixelNeRF [20]	CVPR'21	Once	Fast	+	–	Image-conditioned
IBRNet [21]	CVPR'21	Once	Med	+	–	Cross-view attention
MVSNeRF [22]	ICCV'21	Once	Fast	+	–	Cost volume
D-NeRF [23]	CVPR'21	Slow	Slow	–	+	Deformation field
Nerfies [24]	ICCV'21	Slow	Slow	–	+	Latent deformation
HyperNeRF [25]	TOG'21	Slow	Slow	–	+	Ambient space
K-Planes [26]	CVPR'23	Med	Fast	–	+	4-D feature planes

3 Scene Representations for NVS

3.1 Implicit Representations: NeRF and Variants

Original NeRF [12]. An MLP maps 5-D input (x, y, z, θ, ϕ) to $(\mathbf{c}, \sigma)$. Positional encoding lifts inputs to a high-frequency space. Training minimises photometric loss via Eq. 1. Achieves state-of-the-art synthesis quality but requires per-scene optimisation ($\sim$1–2 days on a single GPU).

Acceleration. Instant-NGP [15] uses a multi-resolution hash grid, reducing training to seconds. TensoRF [16] exploits CP/VM tensor decomposition for fast and compact representation. Plenoxels [17] replaces the MLP with a sparse voxel grid, enabling minute-scale optimisation.

Generalisation. pixelNeRF [20] conditions the network on pixel-aligned CNN features, enabling single-image reconstruction (D2). IBRNet [21] aggregates cross-view features via attention. MVSNeRF [22] builds a cost volume from stereo for geometry-aware conditioning.

Dynamic NeRF. D-NeRF [23] introduces a deformation network mapping each point to canonical space. Nerfies [24] and HyperNeRF [25] model complex non-rigid deformations via latent codes and ambient spaces. K-Planes [26] decomposes 4-D space-time into six feature planes.

Table 2 summarises key NeRF variants.

3.2 Explicit Representations: 3DGS and Variants

3D Gaussian Splatting [27]. Scene represented as anisotropic Gaussian primitives optimised via Eq. 2. Adaptive densification adds/prunes Gaussians during training. Real-time rendering ($\sim$100 fps at 1080p) and high quality.

Dynamic 3DGS. 4DGS [28] extends Gaussians into 4-D space-time. SpacetimeGaussians [29] uses polynomial time trajectories. Deformable3DGS [30] applies a deformation network to static Gaussians.

Feed-Forward Generalisation. pixelSplat [31] predicts Gaussians from image-pair features in a single forward pass (D2). MVSplat [32] extends this with a cost-volume prior. Splatt3R [33] builds on DUSt3R for unposed image pairs.

Compression and Editing. LightGaussian [34] prunes and quantises Gaussians for compact storage. Scaffold-GS [35] anchors Gaussians to a sparse structure.

Table 3 summarises key 3DGS variants.

Table 3: Representative 3DGS variants and their properties.

Method	Venue	Real-time	Dynamic	Generalise	Compress	Key Idea
3DGS [27]	TOG'23	+	–	–	–	Gaussian splatting
4DGS [28]	CVPR'24	+	+	–	–	4-D Gaussians
SpacetimeGS [29]	CVPR'24	+	+	–	–	Polynomial traj.
Deformable3DGS [30]	CVPR'24	+	+	–	–	Deformation net
pixelSplat [31]	CVPR'24	+	–	+	–	Feed-forward pred.
MVSplat [32]	ECCV'24	+	–	+	–	Cost volume
Splatt3R [33]	arXiv'24	+	–	+	–	DUSt3R backbone
LightGaussian [34]	NeurIPS'23	+	–	–	+	Pruning+quant.
Scaffold-GS [35]	CVPR'24	+	–	–	+	Anchor structure

3.3 Hybrid and Structured Representations

Triplane. EG3D [36] encodes a scene on three axis-aligned feature planes, decoded by a shallow MLP. This provides an efficient 3-D-aware latent space for GAN and diffusion models and is widely adopted as a backbone for generative NVS.

Gaussian Opacity Fields. GOF [37] combines Gaussian Splatting with a continuous opacity field, bridging explicit and implicit representations for better surface extraction.

4 Generative Paradigms for NVS

4.1 GAN-Based 3-D-Aware Generation

GRAF [38] and pi-GAN [39] condition a radiance-field generator on a latent code, producing multi-view consistent images. EG3D [36] introduced the triplane backbone and a super-resolution network, dramatically improving quality. These methods established the paradigm of *implicit 3-D supervision* via GAN training on 2-D images only.

4.2 Diffusion-Based NVS

Zero-1-to-3 [41] fine-tunes a latent diffusion model on multi-view data, conditioning on relative pose for single-image NVS. SyncDreamer [42] jointly denoises multiple views with shared 3-D attention. One-2-3-45 [43] lifts diffusion outputs to 3-D. Cat3D [48] and ReconFusion [49] apply multi-view diffusion to sparse-view inputs fused via NeRF/3DGS.

4.3 Score Distillation Sampling (SDS)

DreamFusion [44] optimises a NeRF so that renders score highly under a pre-trained 2-D diffusion model, enabling text-driven 3-D generation without 3-D supervision. Magic3D [45] and ProlificDreamer [46] improve quality via coarse-to-fine and variational score distillation. GaussianDreamer [47] combines SDS with 3DGS for faster convergence.

4.4 Video Diffusion for 4-D NVS

ViewCrafter [50] conditions a video diffusion model on camera trajectories for temporally coherent novel-view sequences. WonderWorld [51] generates interactive 3-D scenes from a single image using flow-based priors. Consistent3D [52] enforces cross-view consistency in video diffusion for 4-D generation.

Table 4 summarises generative NVS methods.

5 From NVS to Generative World Modeling

5.1 Dynamic Scene Modeling and 4-D World Representation

Extending NVS from static scenes to dynamic worlds requires modelling how the scene state evolves over time (desideratum D3). We identify two complementary strategies.

Table 4: Representative generative NVS methods.

Method	Venue	Backbone	Input	3D Consistent	Dynamic	Key Contribution
GRAF [38]	NeurIPS'20	NeRF-GAN	Noise	$\sim$	–	Radiance field GAN
pi-GAN [39]	CVPR'21	NeRF-GAN	Noise	$\sim$	–	Periodic implicit GAN
EG3D [36]	CVPR'22	Triplane	Noise	+	–	Triplane + SR
Zero-1-to-3 [41]	ICCV'23	LDM	1 img	$\sim$	–	Pose-conditioned diff.
SyncDreamer [42]	arXiv'23	LDM	1 img	+	–	Joint multi-view denoise
One-2-3-45 [43]	NeurIPS'23	LDM+recon	1 img	+	–	Diffusion $\rightarrow$ 3-D
DreamFusion [44]	ICLR'23	LDM+NeRF	Text	$\sim$	–	Score distillation (SDS)
Magic3D [45]	CVPR'23	LDM+mesh	Text	$\sim$	–	Coarse-to-fine SDS
ProlificDreamer [46]	NeurIPS'23	LDM+NeRF	Text	$\sim$	–	Variational SDS
GaussianDreamer [47]	CVPR'24	LDM+3DGS	Text	$\sim$	–	SDS + 3DGS init
Cat3D [48]	arXiv'24	MV-Diff	Sparse	+	–	MV diffusion + NeRF
ReconFusion [49]	CVPR'24	MV-Diff	Sparse	+	–	Diffusion priors
ViewCrafter [50]	arXiv'24	Video-Diff	1 img	+	+	Cam-cond. video diff.

Table 5: NVS vs. World Modeling: capability boundary. + required and addressed; $\sim$ partially; – not required/absent.

Capability	NVS (static)	World Model
Viewpoint-consistent rendering (D1)	+	+
Sparse/single-image input (D2)	$\sim$	+
Temporal dynamics (D3)	–	+
Action/agent conditioning (D4)	–	+
Physical plausibility (D5)	–	+
Cross-scene generalisation	$\sim$	+
Causal reasoning	–	+

Deformation-based methods maintain a canonical static representation and learn a time-dependent warp: D-NeRF [23], Nerfies [24], 4DGS [28].

Space-time methods directly parameterise a 4-D function: K-Planes [26] uses six planes including a temporal dimension; SpacetimeGaussians [29] assigns polynomial trajectories to each Gaussian.

5.2 Controllable and Interactive World Generation

The key step from NVS to a full world model is *action conditioning* (D4): the ability to modify scene state in response to agent actions.

Genie 2 [53] generates persistent 3-D environments from a single image and supports interactive first-person navigation. Cosmos [54] scales to physical world simulation via internet-scale video pre-training. WonderJourney [55] generates sequential scene explorations guided by text instructions.

5.3 Embodied AI and Autonomous Driving

UniSim [56] builds a closed-loop neural sensor simulator for urban driving, rendering novel views on demand for policy evaluation. GAIA-1 [57] is a generative world model conditioned on ego-actions, video, and text for diverse driving scenario generation. In robotics, NeRF-based object models have been used for manipulation planning [58] and the spatial consistency of NVS is essential for reliable sim-to-real transfer.

5.4 The Boundary and Connection Between NVS and World Modeling

Table 5 precisely delineates the boundary. NVS enforces viewpoint-conditioned rendering with multi-view geometric consistency. A world model additionally requires temporal consistency, action responsiveness, and causal understanding. NVS is *necessary but not sufficient* for world modeling: it provides the spatial grounding that prevents hallucination, while temporal and action modules extend it to the full world-modeling requirement.

Table 6: Key NVS datasets. "RT" = real, "Syn" = synthetic.

Dataset	Year	Type	Dynamic	Scale	GT Depth	Primary Use
NeRF-Synthetic [12]	2020	Syn	–	Object	+	NeRF eval
LLFF [13]	2019	RT	–	Scene	–	Forward-facing
DTU [61]	2014	RT	–	Object	+	MVS/NVS
Tanks & Temples [60]	2017	RT	–	Large	+	Reconstruction
Mip-NeRF 360 [19]	2022	RT	–	Scene	–	Unbounded
ScanNet++ [62]	2023	RT	–	Indoor	+	Generalisation
D-NeRF [23]	2021	Syn	+	Object	–	Dynamic NeRF
HyperNeRF [25]	2021	RT	+	Object	–	Topological change
Neu3D [65]	2022	RT	+	Human	–	Dynamic 3DGS
Waymo Open [66]	2020	RT	+	Urban	+	Driving NVS
nuScenes [67]	2020	RT	+	Urban	∼	World model

6 Datasets and Benchmarks

6.1 Static Scene Datasets

NeRF-Synthetic [12]: 8 synthetic objects with known ground truth; the standard for controlled NVS evaluation. **LLFF** [13]: 8 forward-facing real scenes. **Tanks and Temples** [60]: large-scale outdoor and indoor with LiDAR ground truth. **DTU** [61]: 124 multi-view stereo scenes with calibrated lighting. **Mip-NeRF 360** [19]: 9 unbounded 360° real scenes. **ScanNet++** [62]: high-resolution indoor scenes with DSLR and iPhone captures.

6.2 Dynamic Scene Datasets

D-NeRF dataset [23]: 6 monocular synthetic dynamic objects. **HyperNeRF dataset** [25]: topologically varying monocular captures. **NHR** [63]: multi-view human performance videos. **CMU Panoptic** [64]: dense multi-view social activity captures. **Neu3D** [65]: multi-view dynamic human videos with complex motions.

6.3 Large-Scale and Autonomous Driving

Waymo Open [66]: multi-camera + LiDAR driving data. **nuScenes** [67]: 360° driving with HD maps. **KITTI-360** [68]: urban driving with panoptic annotations for scene-level NVS.

Table 6 provides a structured comparison.

7 Evaluation and Limitations

7.1 Standard NVS Metrics

PSNR (peak signal-to-noise ratio) measures pixel-level fidelity. **SSIM** [69] captures structural similarity. **LPIPS** [70] measures perceptual distance via deep features and correlates better with human judgement than PSNR.

7.2 Video and World Modelling Metrics

FVD [71] (Fréchet Video Distance) measures distributional quality of generated video sequences and is the standard for temporal coherence. **Task-specific metrics**: for autonomous driving, closed-loop collision rate and route completion; for robotics, task success rate.

7.3 Limitations of Existing Evaluation

Current evaluation protocols are insufficient for world-modelling assessment. PSNR/SSIM reward pixel-level closeness but a model can score high while violating 3-D consistency. FVD captures perceptual quality but not geometric accuracy. There is no standard benchmark jointly evaluating geometry, dynamics, action responsiveness, and physical plausibility—a critical gap.

8 Open Problems and Research Roadmap

8.1 The Geometry–Generation Trade-Off

Reconstruction methods (NeRF, 3DGS) deliver geometric precision but lack generative priors; diffusion methods provide rich priors but can violate multi-view consistency. Closing this gap—*e.g.*, via 3DGS-guided diffusion or diffusion-seeded 3DGS—is the central open problem.

8.2 Long-Horizon Temporal Consistency

Existing 4-D methods maintain coherence over seconds; quality degrades over longer horizons. Solutions require persistent scene memory (*e.g.*, Gaussian maps that accumulate observations) or stronger temporal priors from world-model pre-training on long video.

8.3 Physical Grounding

Current methods learn appearance correlations, not physical laws. Promising directions: integration with differentiable physics simulators [59] and energy-based physical priors.

8.4 Evaluation Methodology

We propose a unified world-modelling benchmark covering: (i) geometric consistency under adversarial viewpoints; (ii) temporal coherence over long horizons; (iii) action responsiveness in interactive settings; (iv) physical plausibility via physics-engine verification.

8.5 Research Roadmap

R1. Unified architectures combining explicit 3-D representations with diffusion-model priors.

R2. Internet-scale feed-forward generalisation for open-world NVS.

R3. Differentiable physics priors integrated into 4-D world models.

R4. Comprehensive world-modelling evaluation suite.

R5. Language and semantic grounding for action-conditioned generation.

9 Conclusion

We have surveyed Novel View Synthesis through the lens of Generative World Modeling, tracing the evolution from classical reconstruction through NeRF, 3DGS, and diffusion-based synthesis toward 4-D world generation. Our unified continuous-spectrum taxonomy organises more than 150 methods along three axes (input regime, temporal scope, controllability) and makes explicit their world-modelling implications. The related-survey comparison (Table 1) demonstrates a clear gap that this work addresses: no existing survey jointly covers all three paradigms with a world-modelling lens.

The field is at an inflection point where spatial grounding from NVS and temporal/action capabilities from world modelling are converging. Systems such as Genie 2, Cosmos, and ViewCrafter already embody this convergence. Closing the geometry–generation trade-off, achieving long-horizon coherence, and grounding predictions in physical laws are the primary challenges ahead.

Draft v0.2 — ongoing work. Content will be expanded in subsequent versions with broader coverage and updated citations.

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